

MATH 4162/Math 6201
Numerical methods for differential equations
Fall 2025

Homework problem set #3

Show all necessary steps of working in the given space

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Upload the assignment solution no later than

23:59 on Nov 17, 2025

Full marks 50

1. [25pts]

(a) Verify that $\{1, \cos x, \cos 2x, \dots\}$ is an orthogonal set on the interval $x \in [-\pi, \pi]$.

(b) Consider

$$\phi_n(x) = e^{2\pi i n x / L}$$

and verify that $\{\phi_n(x)\}$ is orthogonal on the interval $[0, L]$.

(c) The Leap-Frog scheme for

$$\frac{\partial u}{\partial t} + a(u) \frac{\partial u}{\partial x} = 0, \quad a > 0$$

using Fourier collocation method on a grid $\{x_j\}$ was given in the lecture notes. Determine the stability condition and show that

$$\Delta t \leq \frac{\Delta x}{\pi \max a(u)}.$$

(d) Consider a spectral differentiation of a PDE given as

$$\frac{\partial U}{\partial t} = \mathcal{D}U,$$

where $U = [u(x_0), \dots, u(x_{n-1})]^T$ denotes the solution of PDE as n -dimensional vector, and $\mathcal{D} \in \mathbb{R}^{n \times n}$ is the differentiation matrix. Let $E = (1/2)U^T U$ be the energy. Show that the energy is conserved if the differentiation matrix $\mathcal{D}$ is skew-symmetric.

(e) Consider the spectral differentiation matrix $\mathcal{D}$ given in lecture notes (lecture#13, page 4). If the 1D advection equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0, \quad a > 0$$

is discretized using the spectral matrix $\mathcal{D}$, show that the energy will not be conserved for any standard time integration method. Then propose a modification of $\mathcal{D}$ that ensures energy conservation?

a) Orthogonality of $\{1, \cos x, \cos 2x, \dots\}$ on $[-\pi, \pi]$

Inner product: $\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) g(x) dx$

Case 1: Constant with cosine ($n \geq 1$)

$$\begin{aligned}\langle 1, \cos nx \rangle &= \int_{-\pi}^{\pi} \cos nx \, dx \\&= \left[\frac{\sin(nx)}{n} \right]_{-\pi}^{\pi} \\&= \frac{\sin(n\pi)}{n} - \frac{\sin(-n\pi)}{n} = 0\end{aligned}$$

Case 2: Different cosines ($m \neq n$)

$$\langle \cos(mx), \cos(nx) \rangle = \int_{-\pi}^{\pi} \cos(mx) \cos(nx) \, dx$$

We know $\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$

$$\Rightarrow \frac{1}{2} \int_{-\pi}^{\pi} [\cos((m-n)x) + \cos((m+n)x)] \, dx$$

$$\Rightarrow \frac{1}{2} \left[\frac{\sin((m-n)x)}{m-n} + \frac{\sin((m+n)x)}{m+n} \right]_{-\pi}^{\pi}$$

$$\Rightarrow 0$$

Case 3: Same function ($m=n$)

$$\langle \cos(nx), \cos(nx) \rangle = \int_{-\pi}^{\pi} \cos^2(nx) \, dx$$

$$\text{We know } \cos^2 \theta = \frac{1}{2} (1 + \cos(2\theta)) \quad dx = \pi$$

Therefore the set is orthogonal.

$$b.) \quad \phi_n(x) = e^{2\pi i n x/L} \rightarrow [0, L]$$

$$\begin{aligned} \langle \phi_m, \phi_n \rangle &= \int_0^L e^{-2\pi i m x/L} \cdot e^{2\pi i n x/L} dx \\ &= \int_0^L e^{2\pi i (n-m) x/L} dx \end{aligned}$$

Case 1: $m=n$

$$\Rightarrow \int_0^L 1 dx$$

$$\Rightarrow L$$

Case 2: $m \neq n$

$$\Rightarrow \left[\frac{e^{2\pi i (n-m) x/L}}{2\pi i (n-m)/L} \right]_0^L$$

$$\Rightarrow \frac{L}{2\pi i (n-m)} \cdot e^{\underset{\substack{\downarrow \\ 1}}{2\pi i (n-m)}} - 1$$

$$\Rightarrow \frac{L}{2\pi i (n-m)} \cdot [1 - 1] = 0$$

$\therefore \phi_n(x)$ is orthogonal on $[0, L]$

c.) Leap-Frog scheme

$$\frac{(u^{n+1} - u^{n-1}))}{2\Delta t} + a(u) \frac{\partial u}{\partial x} = 0$$

Amplification factor:

$$\varepsilon^2 - 2i\lambda\varepsilon - 1 = 0$$

for stability, $|\varepsilon| \leq 1$.

$$|\lambda| \leq 1$$

For spectral methods, the maximum no
is $k_{\max} \approx \frac{\pi}{\Delta x}$

$$\Rightarrow \left| a(u) \Delta t \cdot \frac{\pi}{\Delta x} \right| \leq 1$$

$$\therefore \Delta t \leq \frac{\Delta x}{\pi \max |a(u)|}$$

d.)

$$\frac{\partial u}{\partial t} = Du$$

$$E = \frac{1}{2} U^T U \quad \longrightarrow \text{energy}$$

$$\begin{aligned} \frac{dE}{dt} &= \frac{d}{dt} \left[\frac{1}{2} U^T U \right] \\ &= \frac{1}{2} \left[\frac{dU^T}{dt} \cdot U + U^T \cdot \frac{dU}{dt} \right] \\ &= \frac{1}{2} \left[(DU)^T U + U^T (DU) \right] \\ &= \frac{1}{2} \left[U^T D^T U + U^T D U \right] \end{aligned}$$

we know $D^T = -D$ $\longrightarrow$ skew symmetric

$$\begin{aligned} &= \frac{1}{2} \left[U^T (-D) U + U^T D U \right] \\ &= \frac{1}{2} \left[\cancel{-U^T D U} + \cancel{U^T D U} \right] \end{aligned}$$

$$= 0$$

$\therefore$ Energy is conserved when D is skew symmetric

e.) 1D advection eqn.

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

Discretized: $\frac{dU}{dt} = -aDU$

$$\Rightarrow \frac{dE}{dt} = U^T \frac{dU}{dt} = -aU^T D U$$

Since $D + D^T \neq 0$, $U^T D U \neq 0$

Here energy is not conserved.

I propose

$$D_{\text{skew}} = \frac{(D - D^T)}{2}$$

2. [25pts]

(a) Consider the linear PDE:

$$\frac{\partial u}{\partial t} + 4 \frac{\partial u}{\partial x} = 0$$

Find the characteristic curves and draw three curves in the $x - t$ plane with starting points $(-5, 0)$, $(0, 0)$, and $(5, 0)$.

(b) Consider the nonlinear PDE:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0, \quad 0 < x < \pi$$

where $u(x, 0) = \sin(x_0)$ with x_0 being the starting location of the characteristic curves passing through (x, t) at $t = 0$.

Find the characteristic curves and draw three curves in the $x - t$ plane with starting points $(1/4, 0)$, $(1/2, 0)$, and $(3/4, 0)$.

Determine if any pair of the curves intersect. Find the point of intersection (x_c, t_c) and/or justify if there is no intersection.

(c) Consider the FV discretization of 1d advection equation. Substituting $Q_j^n = e^{ikj\Delta x}$ show that the amplification factor is

$$G = (1 - C) + Ce^{-ik\Delta x}$$

where $C = a\Delta t/\Delta x$.

(d) Consider the PDE

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = bu, \quad u(x, 0) = u_0(x)$$

Verify that the solution of this PDE is $u(x, t) = e^{bt}u_0(x - at)$. Consider the following scheme

$$Q_j^{n+1} = Q_j^n - \frac{a\Delta t}{\Delta x}(Q_j^n - Q_{j-1}^n) + \Delta t b Q_j^n.$$

Derive the leading order truncation error τ_j^n and show

$$\tau_j^n = \frac{1}{2}\Delta t^2(a^2u_{xx} - 2bau_x + b^2u) + \frac{1}{2}(a\Delta t\Delta x)u_{xx} + \text{higher order terms}$$

Show that the given scheme is Lax-Richtmyer stable provided $a\Delta t/\Delta x \leq 1$.

(e) [Math4162 only] Consider the hyperbolic system of linear PDES:

$$\frac{\partial p}{\partial t} + K_0 \frac{\partial u}{\partial x} = 0$$

$$\frac{\partial u}{\partial t} + \frac{1}{\rho_0} \frac{\partial p}{\partial x} = 0$$

where $K_0 = 4$ and $\rho_0 = 1$. The initial conditions are

$$(p, u)(x, 0) = \begin{cases} (p_l, u_l), & x < 0 \\ (p_r, u_r), & x > 0 \end{cases}$$

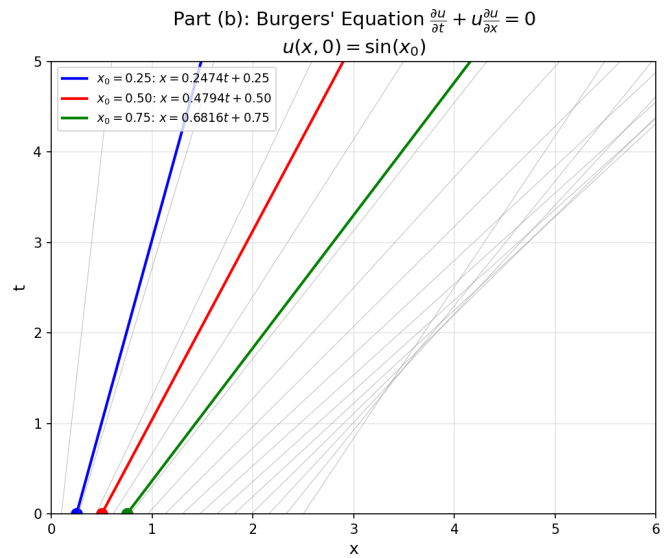
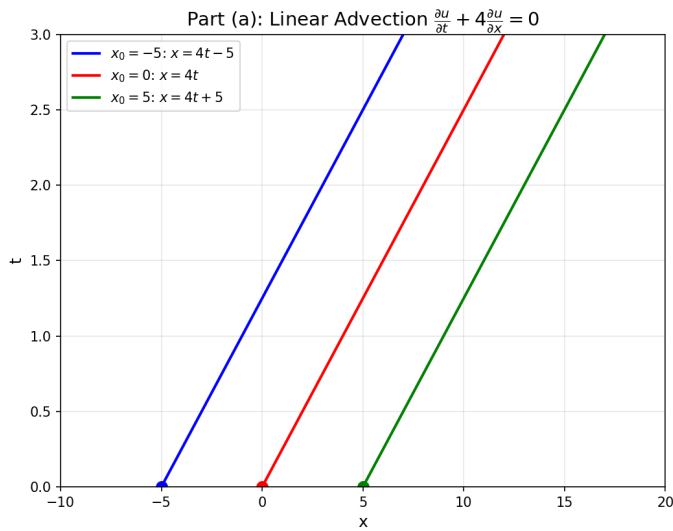
Compute the eigenvalues and (right and left) eigenvectors of the system matrix.

Write the jump $\Delta q = q_r - q_l$ as

$$\Delta q = \alpha_1 \phi_1 + \alpha_2 \phi_2$$

and find the values/expression of α_1 and α_2 .

a.) b.)



c.) $C = \frac{a \Delta t}{\Delta x}$

FV scheme for advection (upwind)

$$Q_j^{n+1} = Q_j^n - C(Q_j^n - Q_{j-1}^n)$$

putting $Q_j^n = e^{ikj\Delta x}$

$$\begin{aligned} Q_j^{n+1} &= e^{ikj\Delta x} - C(e^{ikj\Delta x} - e^{ik(j-1)\Delta x}) \\ &= e^{ikj\Delta x} [1 - C(1 - e^{ik\Delta x})] \end{aligned}$$

Amplification factor

$$G = \frac{Q_j^{n+1}}{Q_j^n} = 1 - C + Ce^{-ik\Delta x}$$

$$\therefore G = (1 - C) + Ce^{-ik\Delta x}$$

d.) PDE: $\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = bu$

Exact soln: $u(x,t) = e^{bt}(x-at)$

$$Q_j^{n+1} = Q_j^n - \frac{a\Delta t}{\Delta x} (Q_j^n - Q_{j-1}^n) + \Delta t \cdot b \cdot Q_j^n \quad -①$$

Error

Time:

$$u(x, t + \Delta t) = u + \Delta t \cdot \frac{du}{dt} + \frac{\Delta t^2}{2} \frac{d^2 u}{dt^2} + O(\Delta t^3) \quad -②$$

Spatial:

$$u(x - \Delta x, t) = u - \Delta x \cdot \frac{du}{dx} + \frac{\Delta x^2}{2} \frac{d^2 u}{dx^2} + O(\Delta x^3) \quad -③$$

Substitute ② and ③ into ①

$$\Rightarrow u + \Delta t \frac{du}{dt} + \frac{\Delta t^2}{2} \frac{d^2 u}{dt^2} = u - \frac{a\Delta t}{\Delta x} \left[u - u + \Delta x \frac{du}{dx} - \frac{\Delta x^2}{2} \frac{d^2 u}{dx^2} \right] + \Delta t \cdot b \cdot u$$

$$\Rightarrow \Delta t \frac{du}{dt} + \frac{\Delta t^2}{2} \frac{d^2 u}{dt^2} = -a \Delta t \cdot \frac{du}{dx} + \frac{a\Delta t \Delta x}{2} \frac{d^2 u}{dx^2} + \Delta t \cdot b \cdot u$$

Dividing by Δt

$$\Rightarrow \frac{du}{dt} + a \frac{du}{dx} - bu = \frac{a \Delta x}{2} \frac{d^2 u}{dx^2} - \frac{\Delta t}{2} \frac{d^2 u}{dt^2}$$

PDE: From $\frac{du}{dt} = -a \frac{du}{dx} + bu$

$$\Rightarrow \frac{d^2 u}{dt^2} = -a \frac{du}{dx dt} + b \frac{du}{dt} = -a \left(-a \frac{du}{dx} + bu \right)$$

$$\Rightarrow \frac{d^2 u}{dt^2} = a^2 \frac{d^2 u}{dx^2} - 2ab \frac{du}{dx} + b^2 u$$

Error

$$\Rightarrow \frac{a \Delta x}{2} \frac{d^2 u}{dx^2} - \frac{\Delta t}{2} \left(a^2 \frac{d^2 u}{dx^2} - 2ab \frac{du}{dx} + b^2 u \right)$$

$$\Rightarrow \frac{1}{2} \left(a \Delta x - a^2 \Delta t \right) \frac{d^2 u}{dx^2} + ab \Delta t \frac{du}{dx} - \frac{b^2 \Delta t}{2} u$$

$$\Rightarrow \frac{1}{2} \Delta t^2 \left(a^2 \frac{d^2 u}{dx^2} - 2ba \frac{du}{dx} + b^2 u \right)$$

$$+ \frac{1}{2} (a \Delta t \Delta x) \frac{d^2 u}{dx^2} + \dots$$

Amplification factor:

$$G = 1 - C(1 - e^{-ik\Delta t} + b\Delta t)$$

With the source term, stability requires

$$|1 - C + Ce^{-ik\Delta x} + b\Delta t| \leq 1$$

(f) [Math6201 only] Consider the hyperbolic system of PDEs:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) = 0$$

$$\frac{\partial \rho u}{\partial t} + \frac{\partial}{\partial x}(\rho u^2 + p) = 0.$$

Find the eigenvalues of the system and determine whether the system is hyperbolic or not.

Show that

$$\frac{du}{dt} \pm \frac{a}{\rho} \frac{d\rho}{dt} = 0 \quad \text{along} \quad \frac{dx}{dt} = u \pm a,$$

where $a^2 = \frac{dp}{d\rho}$.

f.)

$$q = [\rho, \rho u]^T$$

Eqn 1 (Mass)

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} = 0$$

$$\frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} + \rho \frac{\partial u}{\partial x} = 0$$

Eqn 2 (Momentum)

$$\frac{\partial (\rho u)}{\partial t} + \frac{\partial (\rho u^2 + p)}{\partial x} = 0$$

$$\rho \frac{\partial u}{\partial t} + u \frac{\partial \rho}{\partial t} + 2\rho u \frac{\partial u}{\partial x} + u^2 \frac{\partial \rho}{\partial x} + \frac{\partial p}{\partial x} = 0$$

From eqn 1, $\frac{\partial p}{\partial t} = -u \frac{\partial p}{\partial x} - \rho \frac{\partial u}{\partial x}$

$$\Rightarrow \frac{\rho \partial u}{\partial t} + u \left(-u \frac{\partial p}{\partial x} - \rho \frac{\partial u}{\partial x} \right) + 2 \frac{\rho u \partial u}{\partial x} + \frac{u^2 \partial p}{\partial x} + \frac{\partial p}{\partial x} = 0$$

$$\Rightarrow \frac{\rho \partial u}{\partial t} - \frac{u^2 \partial p}{\partial x} - \frac{\rho u \partial u}{\partial x} + 2 \frac{\rho u \partial u}{\partial x} + \frac{u^2 \partial p}{\partial x} + \frac{\partial p}{\partial x} = 0$$

$$\Rightarrow \frac{\rho \partial u}{\partial t} + \frac{\rho u \partial u}{\partial x} + \frac{\partial p}{\partial x} = 0$$

Using chain rule

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \left(\frac{a^2}{\rho} \right) \frac{\partial p}{\partial x} = 0$$

Matrix

$$A = \begin{bmatrix} u & \rho \\ a^2/\rho & u \end{bmatrix}$$

$$\det(A - \lambda I) = (u - \lambda)^2 - \rho \cdot \frac{a^2}{\rho} = 0$$

$$\Rightarrow (u - \lambda)^2 = a^2$$

$$\Rightarrow u - \lambda = \pm a$$

Eigenvalues:

$$\lambda_1 = u + a$$

$$\lambda = u - a$$

system is hyperbolic if

a) all eigenvalues are real

b) there exists a complete set of eigenvectors

the system is hyperbolic.